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Oefeningenreeks 6D nr. 9.1

$$s_n = \sum_{k=1}^{n} \frac{1}{k\,(k+1)}$$
$$\Rightarrow \frac{1}{k\,(k+1)} = \frac{1}{k} - \frac{1}{k+1}$$
$$s_n = \sum_{k=1}^{n} \left(\frac{1}{k} - \frac{1}{k+1}\right)$$
$$= \left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \cdots + \left(\frac{1}{n} - \frac{1}{n+1}\right)$$
$$= 1 - \frac{1}{n+1}$$
$$\lim_{n\to+\infty} s_n = \lim_{n\to+\infty} \frac{n}{n+1} = 1$$

Dus de reeks is convergent en de som is 1.

References